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Market Briefs Pro

Read Less News

Good morning, Pro Briefers!

There's a metal hiding in your soda can, your car door, and the military's newest drones - and it's crucial to creating all three, and thousands of other products.

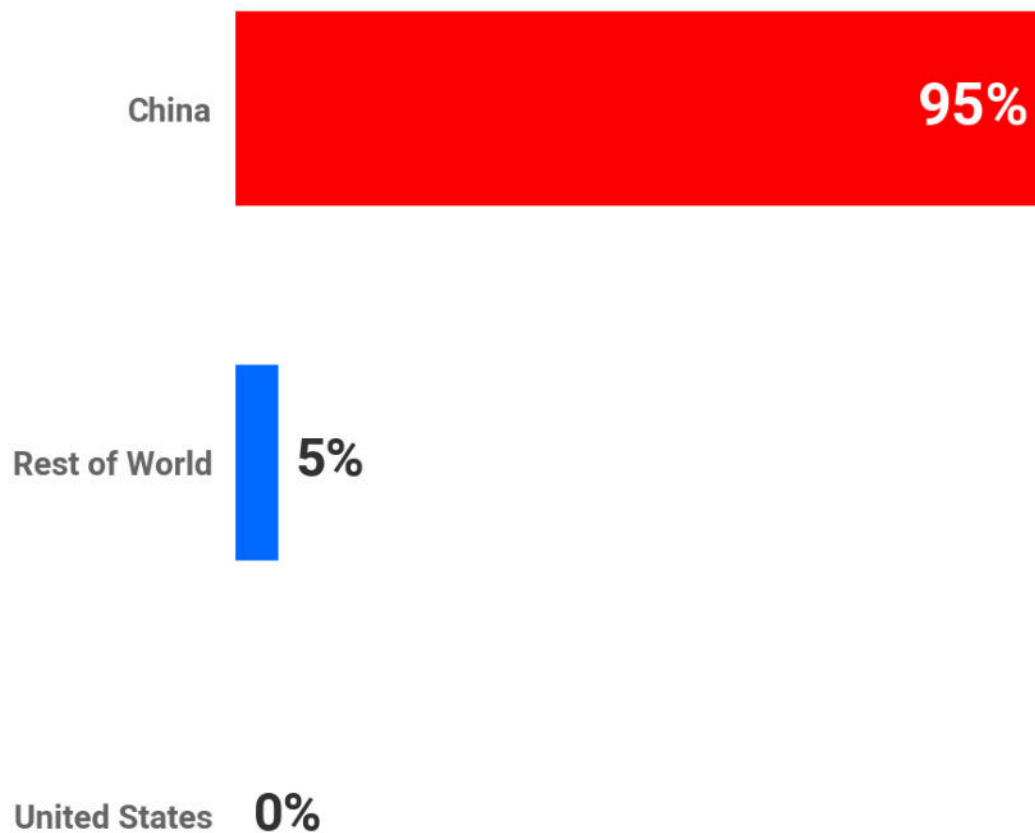
It's called magnesium - but there's a problem: One country produces nearly all of it.

In 2024, China was responsible for 95% of the world's primary magnesium.

That same year? The U.S. made none, which was the third year in a row.

China's Magnesium Stranglehold

Share of global primary magnesium production, 2024



China made 95% of the world's primary magnesium in 2024.

The U.S. made none — for the third year in a row.

Market Briefs **Pro**

Data *via* USGS

But that might be shifting - as the U.S. now looks to be a more dominant player in the magnesium industry, and it's where we have identified a Government Shift:

One country controls the global magnesium market, but new U.S. policies and Western restart projects may flip the picture, opening potential opportunities for investors who pay attention.

Let's break down why investors may want to pay attention to this shift.

Magnesium turns up in a surprising number of places investors should care about:

- It gets blended into aluminum for beverage cans, vehicle body panels, and aircraft parts.
- It's a key ingredient for making titanium, the metal inside fighter jets and submarines.
- It keeps military drones lightweight, helping them fly farther.
- It shows up in helicopter parts where weight savings matter most.

If aluminum, titanium, or defense gear matters to the U.S. economy, magnesium is part of that story.

Where Magnesium Shows Up

**Aluminum alloys — beverage cans,
vehicle body panels, aircraft parts.
~40% of global magnesium ends up here.**

**Titanium production — every kg of
titanium (F-35, Boeing 787, subs)
needs ~1 kg of magnesium via the Kroll
process.**

**Military drones – lightweighting keeps
drones aloft longer; defense
procurement is climbing.**

**Helicopter components – magnesium
alloys go into Black Hawk transmission
parts and future rotorcraft programs.**

**Illumination flares & IR decoys –
Luxfer's Elektron unit supplies
magnesium powders to the U.S.
military.**

Market Briefs [Pro](#)

Data [via](#) USGS

And while China is the biggest producer of magnesium, it's really one major city in China doing the work.

That city is Yulin in northern Shaanxi, but in late 2021, Beijing ordered Yulin smelters to cut output in half.

Remember: magnesium is in thousands of different products - but cutting supply meant prices shot up.

- Magnesium prices jumped more than 230% in the last few months.

Several aerospace and auto supply chains came dangerously close to empty before output recovered.

The bottom line: If one country controls most of the magnesium supply, that puts U.S. interests at risk.

So, the U.S. is now making moves to change that - that's leading to money moving and potential opportunities for investors to profit.

ANALYST TAKE

Tickers we're tracking:

Mining: TTI

Supplying: LXFR

Down-stream: CSTM

THE SHIFT

Why It Matters Now

China has the ability to cut off supply whenever it wants to - that gives it control the U.S. does not have.

So the U.S. has been working to fix that:

- A March 2025 executive order pushed the government to ramp up American mineral production.
- The One Big Beautiful Bill Act (OBBBA) carved out \$2 billion to build up critical mineral stockpiles.
- A program called Project Vault is committing up to \$12 billion toward a strategic mineral reserve.
- Bilateral deals with Japan and Australia in October 2025 set the stage for coordinated supply chain action.

There's also the Defense Logistics Agency, which creates stockpiles of critical materials.

U.S. Critical Mineral Moves

MAR 2025 — Executive Order directs the

federal government to ramp up American mineral production.

OBBBA (2025) – One Big Beautiful Bill Act carves out \$2B for critical mineral stockpile build.

PROJECT VAULT – up to \$12B committed toward a U.S. strategic mineral reserve.

OCT 2025 – Bilateral supply-chain deals signed with Japan and Australia.

Market Briefs [Pro](#)

Data [via](#) The White House

As of early 2026, its priority list covers metals like lithium, nickel, and chromium - but magnesium is missing from it.

That missing piece is where the opportunity sits. If federal money starts pointing at magnesium directly, U.S. producers get a guaranteed price floor that Chinese sellers can't easily beat by cutting prices.

Why does that matter for investors? Western magnesium projects need stronger pricing than Chinese smelters to make money.

Without federal demand, China can drop prices and starve American restart projects of cash.

With a federal floor, U.S. producers gain a reliable customer.

But these stockpiles need to be built - and in the meantime, some companies stand to directly benefit from the Government Shift.

Let's discuss a few below that investors may want to pay attention to.

MINING

Mining Magnesium

If you want public-market exposure to a brand-new American magnesium project, TETRA Technologies (TTI) is really the only option out there.

Here's the story: In December 2025, TETRA signed a joint venture deal with a private startup called Magrathea Metals.

Together, they're working to extract magnesium out of saltwater (called brine) at TETRA's site in Arkansas.

In April 2026, the pilot plant produced clean magnesium metal from the Arkansas brine for the first time, proving the chemistry works.

- Magrathea landed a \$19.6 million Department of Defense award in 2024 to help kickstart American magnesium production.

TETRA is the publicly traded company attached to that work.

TETRA Technologies

Q1 2026 revenue: highest in over a decade

\$156M

Q1 2026 revenue (NYSE: TTI)

THREE REVENUE STREAMS, ONE ARKANSAS SITE

BROMINE

Pays the bills today

Active revenue

Chemical used in oil & gas operations.
Long-running cash-flow business.

MAGNESIUM

The growth engine

Pilot validated Apr 2026

JV with Magrathea (\$19.6M DoD-backed).
Brine-to-Mg pilot succeeded at Arkansas site.
Commercial scale targeted 2027-2028.

LITHIUM

Optionality

Royalty exposure

Royalty potential at the same Arkansas property.
Not yet contributing to revenue.

Market Briefs Pro

Data via TETRA Investor Relations

Magnesium isn't TETRA's only business - the company also makes bromine, a chemical used in oil and gas operations, and has potential lithium royalty exposure at the same property.

That gives the business three potential revenue streams at one site, with bromine already paying the bills today.

We've discussed in previous reports the importance of other minerals, with lithium sitting near the top of that list.

TETRA pulled in \$156 million in revenue during Q1 of 2026, their highest of the last decade.

A huge magnesium project and exposure to other critical minerals give TETRA opportunities to grow in the future and is a company investors will want to keep an eye on.

TETRA Technologies, Inc. (TTI)

YTD Price

\$10.55
(12.59%)



 **Briefs Terminal**

Data [via](#) Briefs Terminal

Exit trigger: A commercial-scale plant likely won't be running until 2027 or 2028 at the soonest.

If the pilot runs into engineering problems while scaling up, or if bromine pricing weakens, that could hurt returns for investors.

And today, bromine income pays for the company's business - magnesium is its potential growth engine, but nothing is proven yet.

Supplying Magnesium

Creating magnesium is just one part of the story - governments and businesses need it today in order to sustain supply chains.

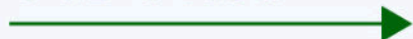
Luxfer Holdings (LXFR) is the only publicly traded U.S. company built around supplying magnesium products to the military.

Its Elektron business makes magnesium powders that go into things like illumination flares and infrared decoys used by U.S. forces.

It also produces specialty magnesium alloys for parts inside Army helicopters, including transmission components on Black Hawk fleets and future rotorcraft programs.

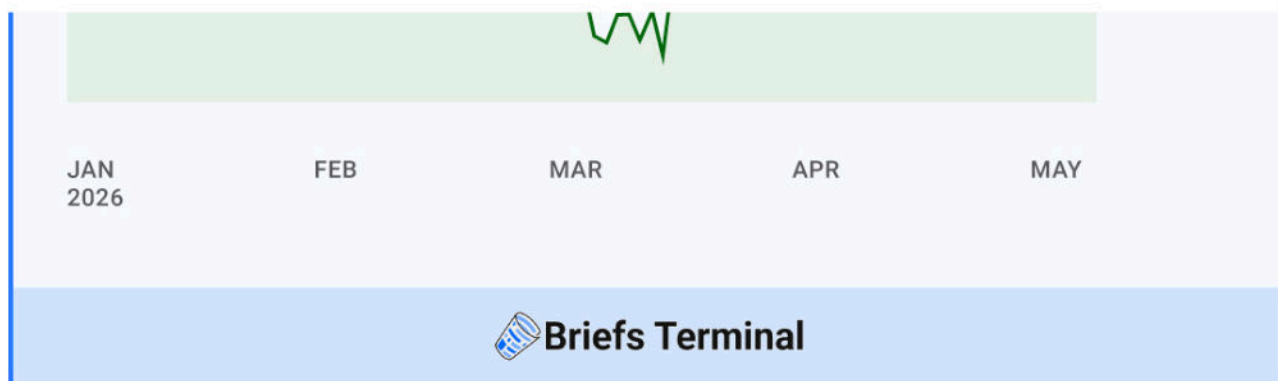
Luxfer Holdings PLC (LXFR)

YTD Price



\$15.56
(15.00%)





Data [via](#) Briefs Terminal

In English: When the Pentagon needs magnesium-based components, Luxfer is often the supplier.

Having a major customer be the U.S. military creates a steady stream of income for Luxfer.

That's because the U.S. government outsources most of its military needs - and these contracts can sometimes last for years and the military budget continues to rise.

Luxfer brought in \$384.6 million in net sales for 2025, with adjusted EPS jumping 12% year over year.

- The company also produced \$26 million in free cash flow with very manageable debt.

Exit trigger: Total sales were slightly down in 2025 even though earnings rose. That signals the gains are coming from improved margins rather than more units sold.

However, if autonomous drone programs or helicopter modernization budgets get scaled back, the steady demand supporting Luxfer's defense magnesium business weakens.

DOWN-STREAM

Down-Stream Play

Fun fact: More than 40% of all magnesium consumed globally ends up alloyed into aluminum products.

Constellium (CSTM) produces aluminum body sheets used by BMW for its 5 Series and similar premium models.

- It makes magnesium-containing aluminum alloys that show up in vehicle doors, roofs, hoods, and structural panels.

Each ton of those alloys can contain anywhere from 30 to 45 kilograms of magnesium.

In July 2024, Constellium landed a \$23 million federal award through the Defense Production Act to expand aluminum casting at its Alabama plant.

Constellium SE (CSTM)

YTD Price



\$32.17
(70.66%)



It's one of the rare cases where critical minerals money has reached an aluminum producer directly - positioning Constellium to benefit from the same policy wave lifting direct magnesium plays.

Keep in mind: Aluminum is a tough business.

Margins get pressured by raw aluminum prices, foreign competition, and energy costs, which means magnesium is only one ingredient in a much bigger cost picture.

Exit trigger: If global aluminum prices fall or European EV demand drops further, Constellium's earnings will react well before any magnesium story plays out.

RISKS

How This Shift Could Change

China is still the dominant supplier by a wide margin. Even in the best case, where all four Western restart projects succeed, U.S. capacity by 2032 may only cover a quarter to 40% of domestic demand.

In short - American dependence on Chinese magnesium will continue for years.

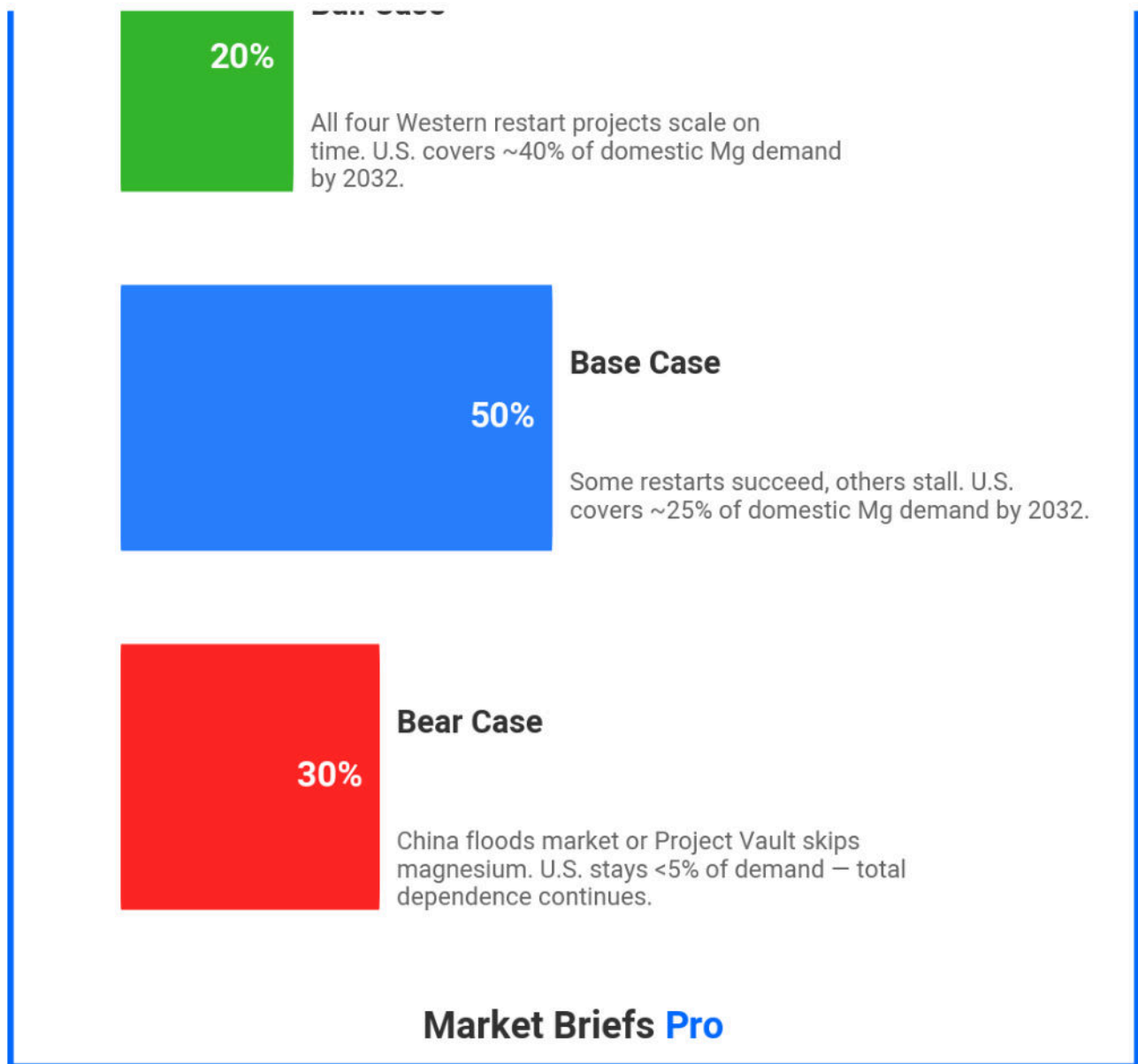
In addition, China also has the power to crush prices on purpose.

If China decides to flood the market, prices can drop low enough to bankrupt U.S. restart projects before they reach scale.

Timing is another problem. Most Western restart projects have no revenue yet. They need either guaranteed federal contracts or a market price spike to make the economics work.

U.S. Magnesium Self-Sufficiency by 2032





There's also a real chance U.S. policy never lands on magnesium specifically.

Washington has the legal authority to act but hasn't aimed those tools at this metal.

If Project Vault funds get spent on lithium, rare earths, or cobalt instead, magnesium stays exactly where it is today.

Finally, the new production methods are unproven at full commercial scale.

Pulling magnesium from brine, seawater, or industrial waste works in small pilot plants. Whether it works at full volumes and competitive prices is still an open question.

That means that some of the adjacent opportunities in this shift could be the most at risk, as their business relies on magnesium.

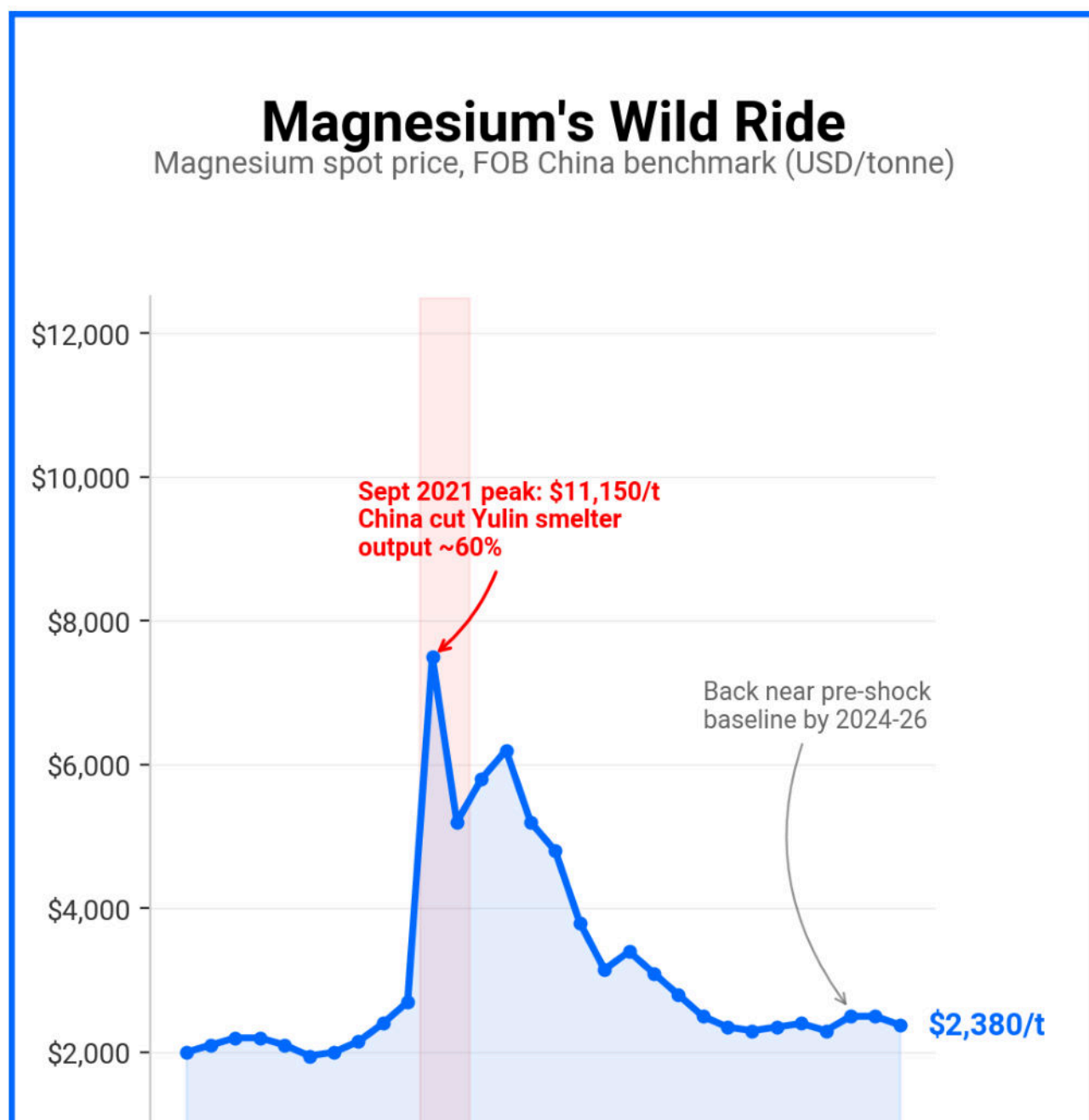
The Metal Shift That Could Move Markets

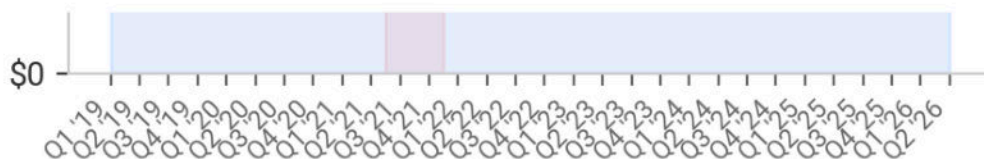
This is a long term shift - but one that could have big implications for the future of our economy one day.

But today, China owns the magnesium market.

The U.S. is working to build alternatives, but those will take years to build.

Still, investors can benefit while this new production is being built by the companies supplying and creating magnesium here at home.





Market Briefs Pro

Data [via](#) USGS

The clearest sign things are shifting will be whether the federal government places magnesium on its priority stockpile list in the coming budget cycle.

Until then, the market remains fragile and could change any day.

Either way, investors may want to keep this shift in mind - as magnesium could one day reshape our industrial economy and the thousands of products that contain magnesium.



- Market Briefs Pro Team

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